

NumPy & Pandas Lab Programs

1. Student Average

Aim

To calculate the average marks of each subject and identify the subject with the highest average.

Algorithm

1. Start
2. Read Students_data.csv
3. Extract subject marks
4. Compute average of each subject
5. Find highest average subject
6. Display results
7. Stop

Code

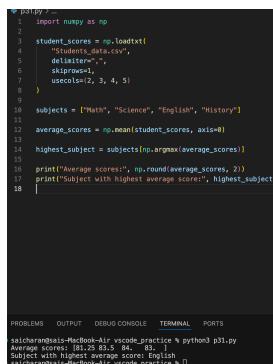
```
import numpy as np

student_scores = np.loadtxt(
    "Students_data.csv",
    delimiter=",",
    skiprows=1,
    usecols=(2, 3, 4, 5)
)

subjects=["Math","Science","English","History"]
average_scores=np.mean(student_scores,axis=0)
highest_subject=subjects[np.argmax(average_scores)]

print("Average scores:",np.round(average_scores,2))
print("Subject with highest average score:",highest_subject)
```

Output



```
> python3 ...
1 import numpy as np
2
3 student_scores = np.loadtxt(
4     "Students_data.csv",
5     delimiter=",",
6     skiprows=1,
7     usecols=(2, 3, 4, 5)
8 )
9
10 subjects = ["Math", "Science", "English", "History"]
11
12 average_scores = np.mean(student_scores, axis=0)
13
14 highest_subject = subjects[np.argmax(average_scores)]
15
16 print("Average scores:", np.round(average_scores, 2))
17 print("Subject with highest average score:", highest_subject)
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
aiicharan@aiicharan-MacBook-Air:~/vscode_practice % python3 p31.py
Average scores: [81.25 83.5 86. 82.]
Subject with highest average score: English
aiicharan@aiicharan-MacBook-Air:~/vscode_practice %
```

Result

Thus, the average marks of each subject were calculated successfully and the highest average subject was identified.

2. Past Month Price Average

Aim

To calculate the average price for the past month using Pandas.

Algorithm

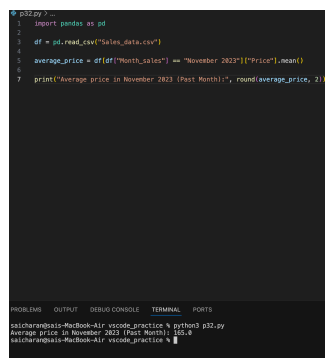
1. Start
2. Read Sales_data.csv
3. Filter November 2023 records
4. Calculate average price
5. Display result
6. Stop

Code

```
import pandas as pd

df=pd.read_csv("Sales_data.csv")
average_price=df[df["Month_sales"]=="November 2023"]["Price"].mean()
print("Average price:",round(average_price,2))
```

Output



```
1 import pandas as pd
2
3 df = pd.read_csv("Sales_data.csv")
4
5 average_price = df[df["Month_sales"] == "November 2023"]["Price"].mean()
6
7 print("Average price in November 2023 (Past Month):", round(average_price, 2))
```

Module: OUTPUT: Jupyter Console: TERMINAL: PLOTS:

saicharan@saicharan-MacBook-Air:~/vscode_projects/Python/pj2.py\$ python pj2.py

Average Price in November 2023 (Past Month): 165.0

saicharan@saicharan-MacBook-Air:~/vscode_projects/Python/pj2.py\$

Result

Thus, the average price for the selected month was calculated successfully.

3. House Average Price

Aim

To calculate the average price of houses having more than 4 bedrooms.

Algorithm

1. Start
2. Read House_data.csv
3. Select houses with bedrooms > 4
4. Calculate average price
5. Display result
6. Stop

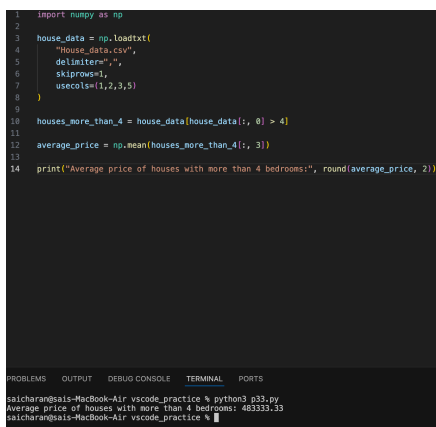
Code

```
import numpy as np

house_data=np.loadtxt("House_data.csv",
delimiter=",",
skiprows=1,
usecols=(1,2,3,5))

houses=house_data[house_data[:,0]>4]
average_price=np.mean(houses[:,3])
print("Average price:",round(average_price,2))
```

Output

A screenshot of a code editor with a dark background. The top part shows the Python code for calculating the average price of houses with more than 4 bedrooms. The bottom part shows the output of the code, which is 'Average price of houses with more than 4 bedrooms: 483333.33'. The code is as follows:

```
1 import numpy as np
2
3 house_data = np.loadtxt(
4     "House_data.csv",
5     delimiter=",",
6     skiprows=1,
7     usecols=(1,2,3,5))
8
9
10 houses_more_than_4 = house_data[house_data[:, 0] > 4]
11
12 average_price = np.mean(houses_more_than_4[:, 3])
13
14 print("Average price of houses with more than 4 bedrooms:", round(average_price, 2))
```

The output is shown in a terminal window at the bottom:

```
saicharan@saicharan-MacBook-Air: ~/vscode_practice % python3 p33.py
Average price of houses with more than 4 bedrooms: 483333.33
saicharan@saicharan-MacBook-Air: ~/vscode_practice %
```

Result

Thus, the average price of houses with more than four bedrooms was calculated successfully.

4. Quarterly Sales Analysis

Aim

To calculate the total yearly sales and percentage increase from Q1 to Q4.

Algorithm

1. Start
2. Read Sales_data.csv
3. Extract month and sales columns
4. Calculate Q1 and Q4 sales
5. Calculate yearly sales
6. Calculate percentage increase
7. Display results
8. Stop

Code

```
import numpy as np

sales_data=np.genfromtxt("Sales_data.csv",
delimiter=",",
skip_header=1,
dtype=str)

months=sales_data[:,1]
sales=sales_data[:,4].astype(float)

Q1=0
Q4=0

for i in range(len(months)):
    if months[i] in ["January","February","March"]:
        Q1+=sales[i]
    elif months[i] in ["October","November","December"]:
        Q4+=sales[i]

total=np.sum(sales)
increase=((Q4-Q1)/Q1)*100

print("Total sales:",total)
print("Percentage increase:",round(increase,2),"%")
```

Output

```
1 import numpy as np
2 import os
3
4 print("Current folder:", os.getcwd())
5 print("Files in this folder:", os.listdir())
6
7 sales_data = np.genfromtxt(
8     "Sales_data.csv",
9     delimiter=";",
10    skip_header=1,
11    dtype=str
12 )
13
14 print("Data read from CSV:")
15 print(sales_data)
16 print("Shape:", sales_data.shape)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

saicharanga@saicharanga-MacBook-Air: ~/Documents/vscode_practice % python3 p34.py

Current folder: /Users/saicharanga/Documents/vscode_practice

Files in this folder: ['hello.py', 'p5.py', 'p1.py', 'Sales_data.csv', 'Sales_data.csv', '.DS_Store', 'House_data.csv', 'p4.py', 'p18.py', 'p24.py', 'p34.py', 'p28.py', 'p14.py', 'p38.py', 'p21.py', 'p15.py', 'p31.py', 'p11.py', 'p25.py', 'p22.py', 'p16.py', 'p32.py', 'p8.py', 'p12.py', 'p26.py', 'p13.py', 'p7.py', 'p23.py', 'p17.py', 'p33.py', 'p8.py', 'p28.py', 'p3.py', 'Students_data.csv', 'p18.py', '.vscode', 'p7.py', 'p18.py', 'p6.py', 'p29.py', 'p2.py']

Data read from CSV:

```
[['October 2023' '120']
 ['November 2023' '150']
 ['November 2023' '120']
 ['December 2023' '200']]
```

Shape: (4, 2)

Result

Thus, the yearly sales and percentage increase from Q1 to Q4 were calculated successfully.